Routing (RIP & OSPF)

1. Which algorithm does RIP and OSPF use?

2. What type of routing protocol is RIP? OSPF?

3. What protocols and ports do RIP and OSPF use?

4. What is the difference between distance-vector and link-state routing?

5. What is the maximum hop count in RIP?

6. How does RIP update its routing table?

7. How does OSPF calculate the shortest path?

8. How does RIP prevent routing loops?

9. How does OSPF prevent routing loops?

10. Explain RIP v1 vs RIP v2 differences and drawbacks+advantages

11. For a large network and a small network, which one should use RIP and which one should use OSPF?

12. Explain the count-to-infinity problem in RIP. Create an example scenario.

13. How does OSPF avoid the count-to-infinity problem?

14. Compare RIP and OSPF based on metric, convergence speed, scalability, and network size. Given two networks, decide which uses RIP and which uses OSPF.

NAT

15. What is NAT and why is it used?

16. How many types of NAT exist?

17. How does Static NAT work?

18. How does Dynamic NAT work?

19. How does PAT work?Explain how NAT/PAT works in a network with multiple private hosts sharing one public IP.

20. Difference between NAT and PAT.

21. Difference between Static and Dynamic NAT.

22. How does NAT assign addresses dynamically?

23. What happens if the NAT translation table is full?

24. Advantages and disadvantages of NAT.

DHCP

25. DHCP? Why used? Which OSI layer does DHCP operate in?

26. How does DHCP assign IP addresses? Explain how DHCP interacts with NAT in a LAN-WAN scenario

27. Explain the DORA process.

28. How does DHCP lease renewal work?

29. Can a device with a static IP communicate in a DHCP network?

30. What happens if two DHCP servers are present in the same network?

31. Configure a DHCP server using both a server and a router? (Practice in Cisco)

IP Addressing & Subnetting

32. How do you find the network and broadcast address of a given IP/CIDR? examples: /16, /18, /23, /25, /26, /27, /28

33. Calculate the number of hosts in a given subnet (e.g., /28, /30, /26, /16, /18, /23, /25).

34. For n hosts, how do you find the required subnet mask and the network addresses? Give an example.

35. For n subnets, how do you find the subnet mask and the network addresses?example.

36. Given a /24 network, calculate maximum subnets possible.

37. Given a /16 network, calculate maximum subnets possible.

38. Divide a /24 network into 4 subnets and write their network addresses.

39. Explain the use of /31 subnet.

40. Explain subnetting formula: 2^n subnets and $2^h – 2$ hosts.

Switching & VLANs

41. What is a VLAN and why is it used?

42. Difference between access port and trunk port.

43. How do you assign a port to a VLAN?

44. How do VLANs reduce broadcast domains?

45. How do two PCs in different VLANs communicate?

46. Configure VLANs based on a topology. (Practice in Cisco)

Scenario Questions

47. If two switches are connected, can a PC communicate with another PC? Explain.

48. Given an IP address and CIDR, find the network, broadcast, and usable hosts.

49. Given a topology, plan subnets and assign them to VLANs.

50. Create a scenario where Dynamic NAT fails to assign IP.